Collaborator or Assistant?

How AI Coding Agents Partition Work Across Pull Request Lifecycles

AI coding agents now open pull requests at scale. When their code merges into the trunk — did anyone actually review it?

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Our contribution: a map of where to look

Raw PR metadata says what happened — not who authorized it, or who answers for it. We turn it into a map.

1 **Each PR is a lifecycle**

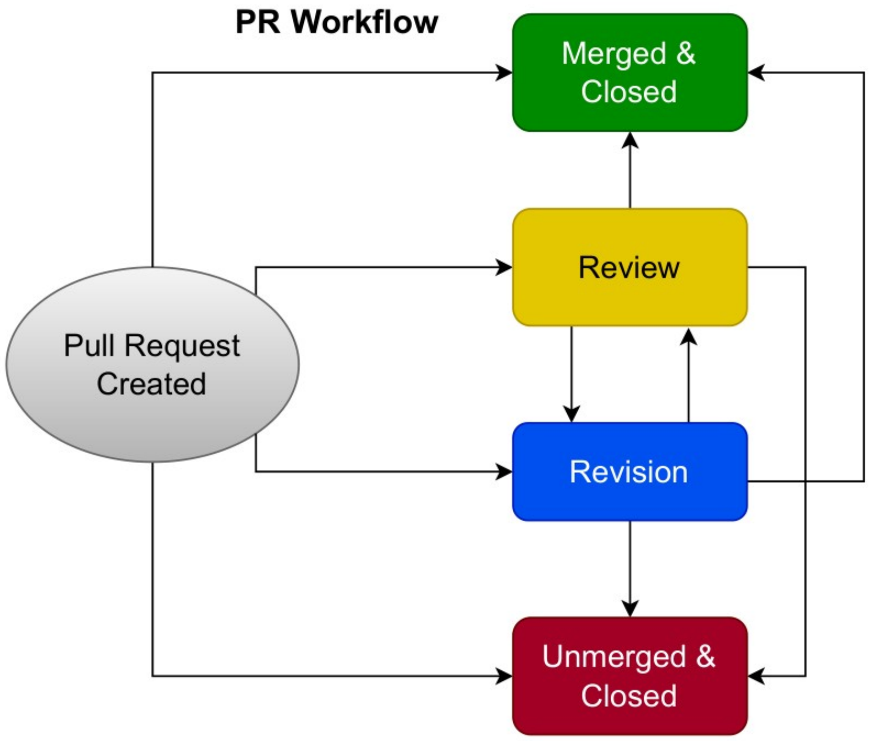
A process to characterize — not a row in a table.

2 **Timeline maps the pathways**

State-machine transitions recover the routes a PR can take.

3 **Three separable observables**

Attribution · review routing · merge — who initiates vs who executes the merge.



The paper's PR-lifecycle state machine (the map).

Same data, a different lens

No new dataset and no new sample — the same PRs and the same metadata. Read the review-and-merge layer (not just initiation) and the same structures tell a governance story.

1. **Construct correction**

   Every PR is agent-associated by construction — 'Human-Init' is an attribution appearance (login-or-name), not human authorship.

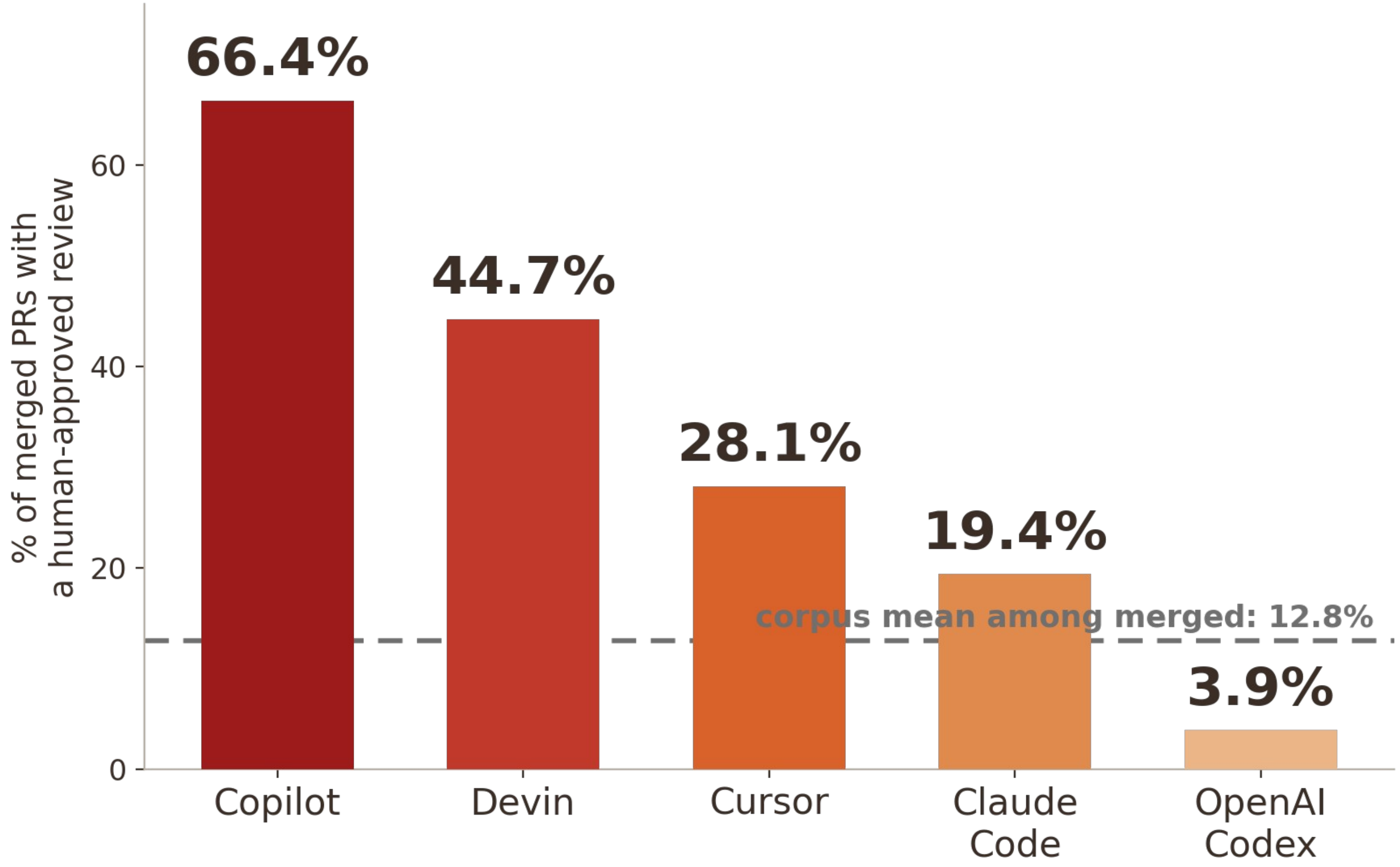
2. **The pathways generalize**

   Same map, fuller data: the Collaborator / Assistant routing profile re-reads through recorded review-and-merge traces.

3. **Reframe, not retract**

   The review axis alone reproduces the split; the small-sample case is caveated.

The twist: recorded human review is the exception



89.6%

of 29,585 PRs reach a terminal state with no recorded human-approved review

17×

across the Collaborator→Assistant spectrum (Copilot 66.4% → Codex 3.9%)

Most agent PRs merge with no recorded human-approved review.

Collaborator or Assistant? A junior teammate — see the poster

Agents do the operational work (≥96% of Collaborator-tool PRs are agent-initiated) yet humans execute almost every merge (<0.1% agent-executed): operational agency and governance authority decouple.

- **The per-tool governance profile —** behind the 17-fold.
- **The construct correction —** attribution as appearance, not identity.
- **Trace vs governance legibility —** what a recorded merge can and cannot evidence.
- **A future-work prototype —** quality / outcome analysis on live AIDev data.

Come to the poster — the instrument, and what it reveals.

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